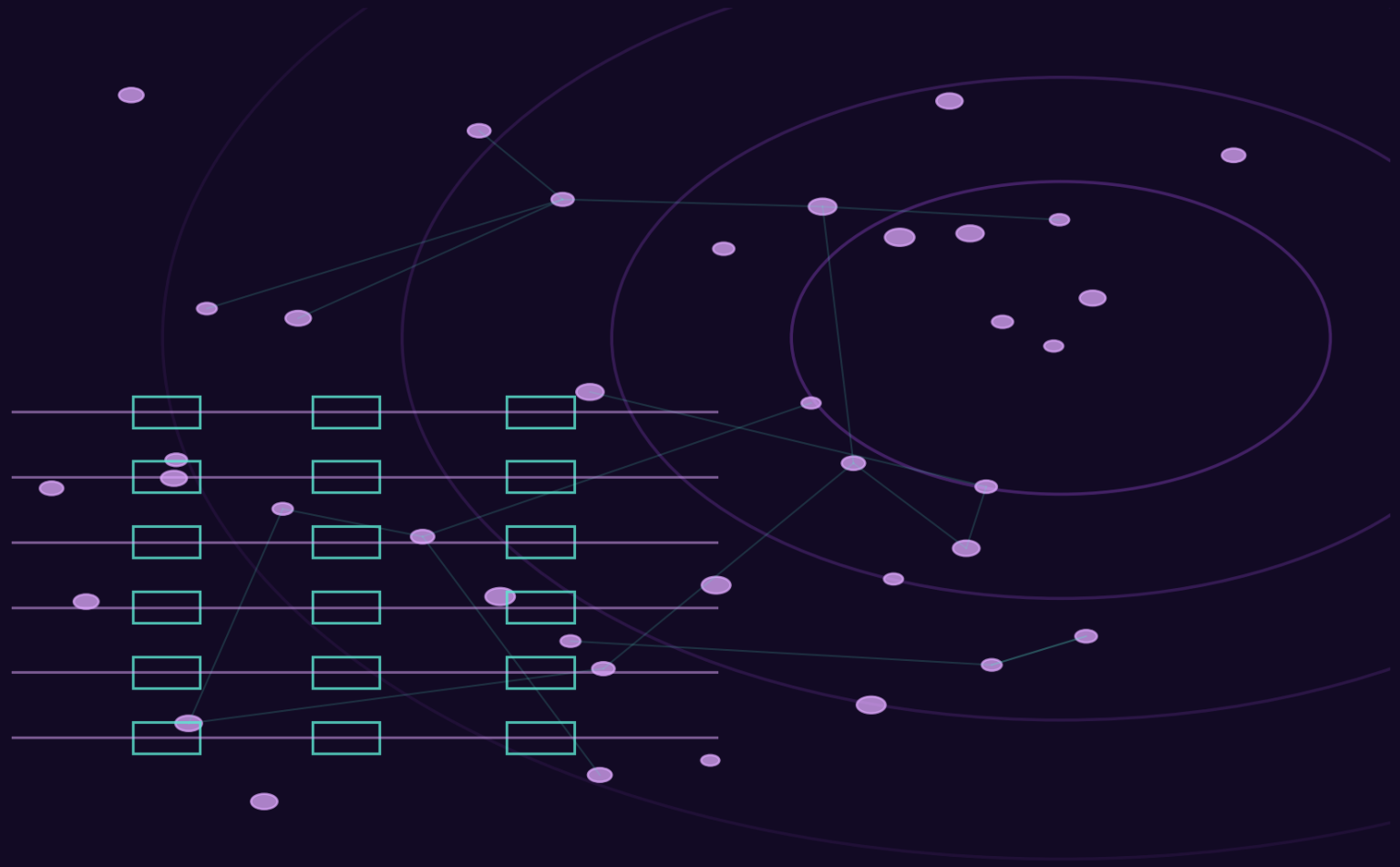


FINTECH · QISKIT · HYBRID QML · QAOA

CertiQ-Chirp

Certified Quantum Campaign Intelligence — one probability, one decision, one measurable outcome.

THE PROBLEM

A bank campaign is one connected decision

Predict

300 labelled customers → calibrated probabilities for 100 hidden-label customers

Select

Choose exactly 6 of the top 10 under budget and capacity limits

Simulate

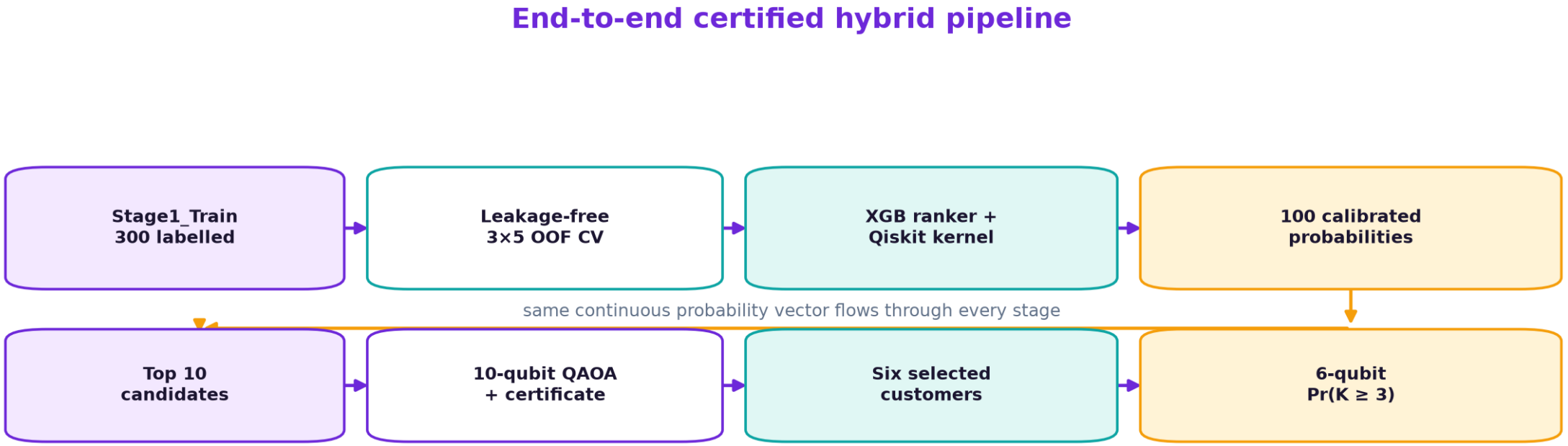
Estimate Pr(K ≥ 3) with one qubit per selected customer

Final answers: six customer IDs and the probability that at least three subscribe.

Quantum components have clear jobs; classical exact references make every result auditable.

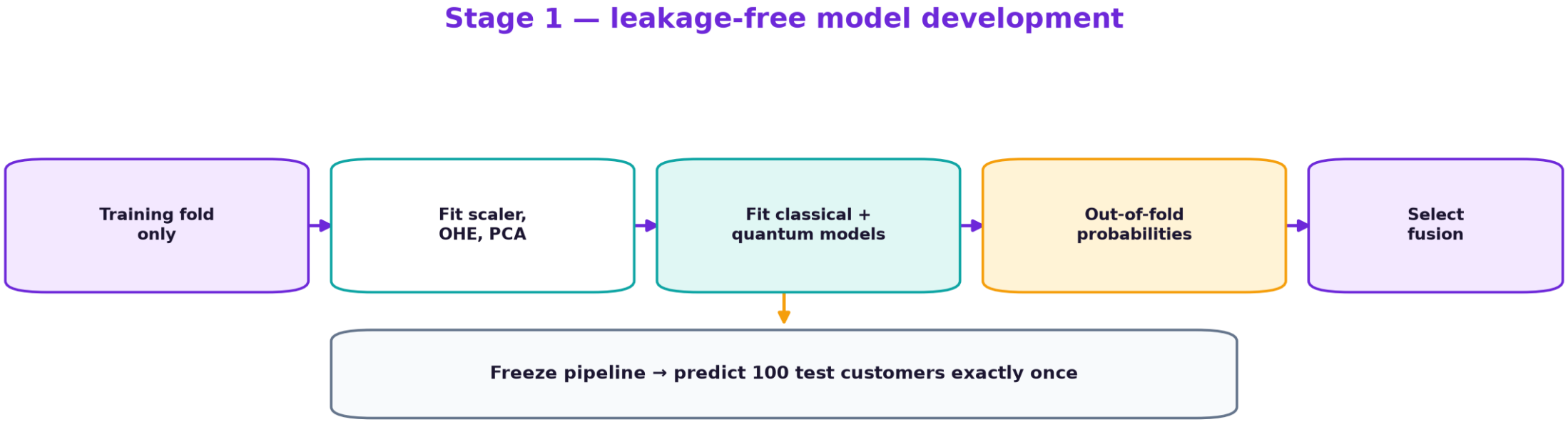
ARCHITECTURE

One traceable hybrid pipeline



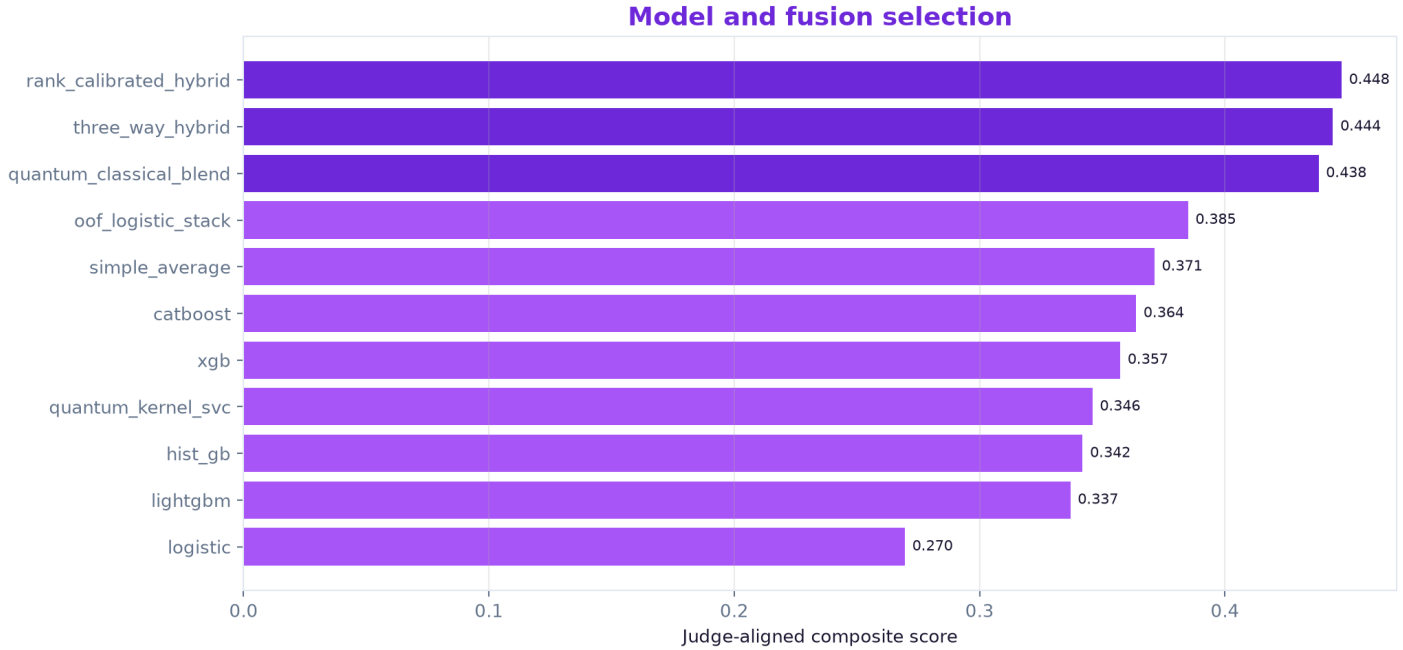
STAGE 1

Leakage-free model development

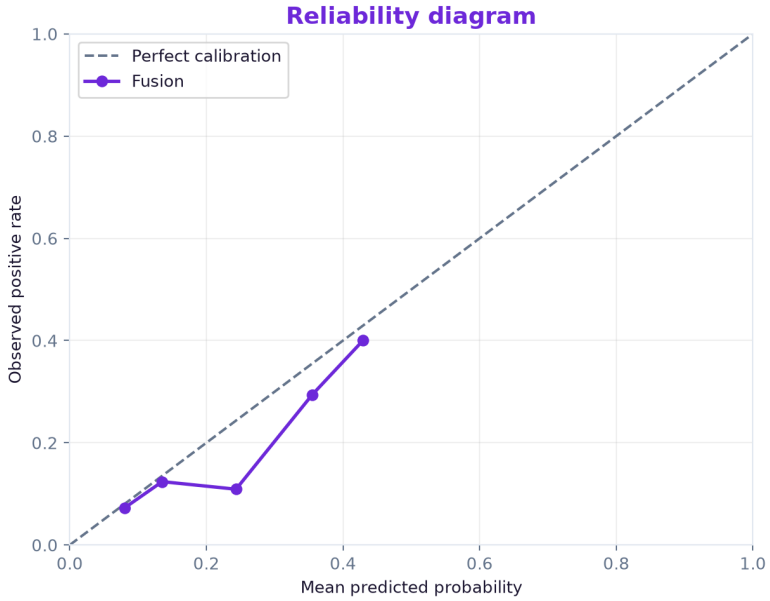


STAGE 1

The quantum kernel is the calibrator



Model selection + reliability

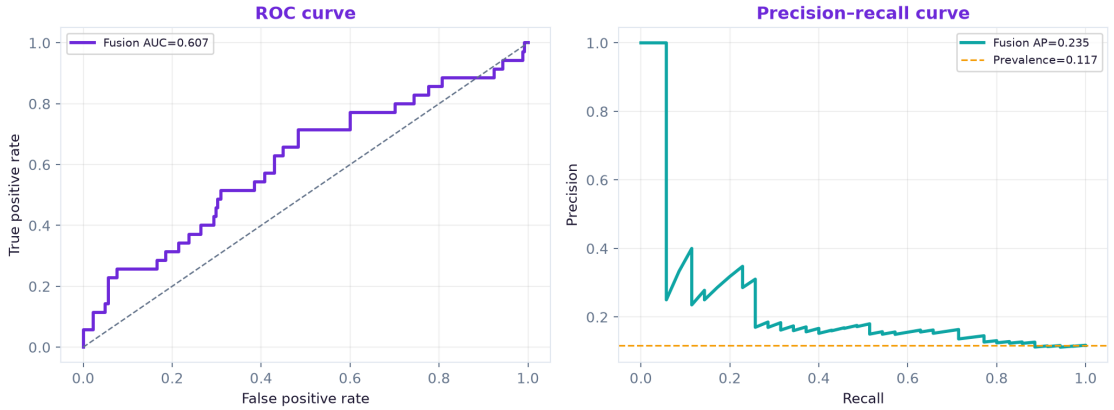


Fusion improves Brier, log loss, and ECE while preserving top-decile

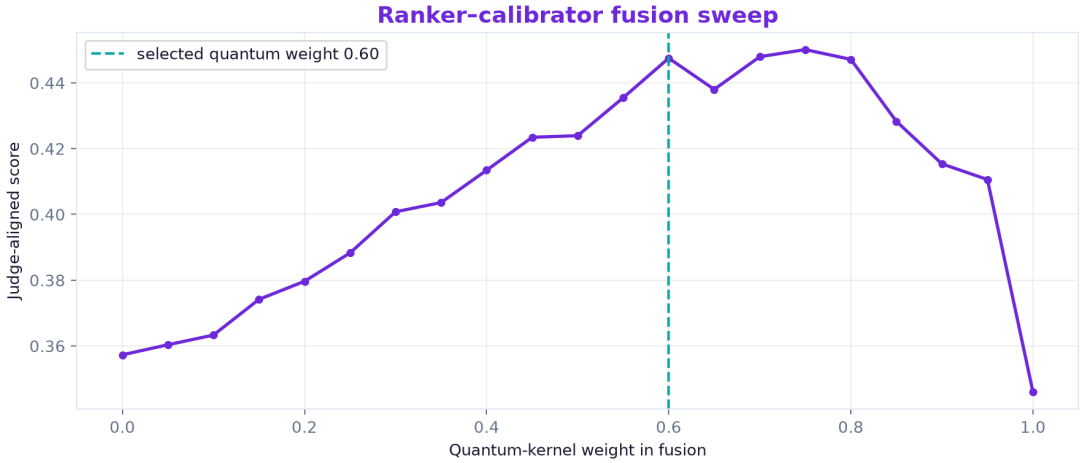
STAGE 1

Ranking quality and weight sensitivity

ROC / PR curves

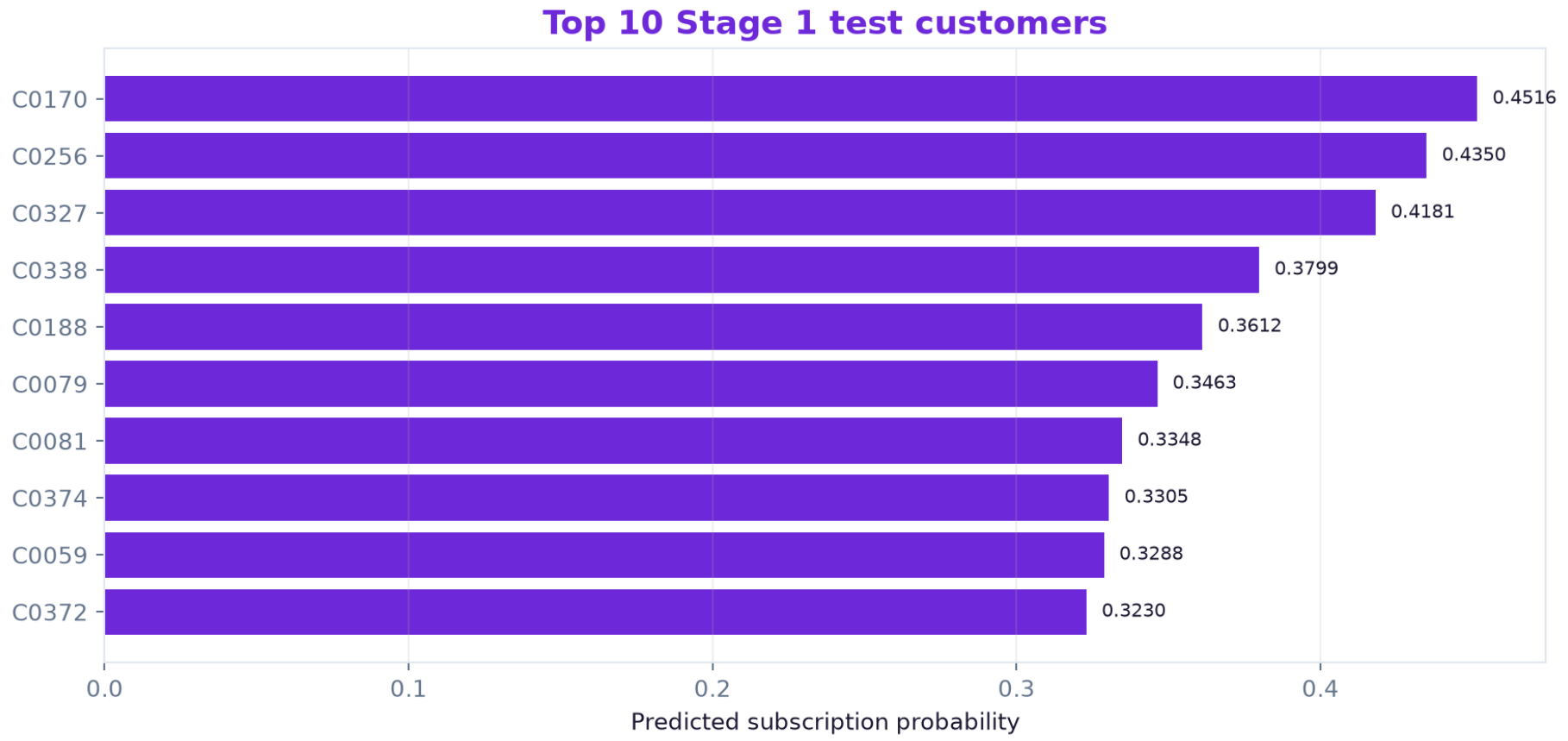


Fusion sweep



STAGE 1

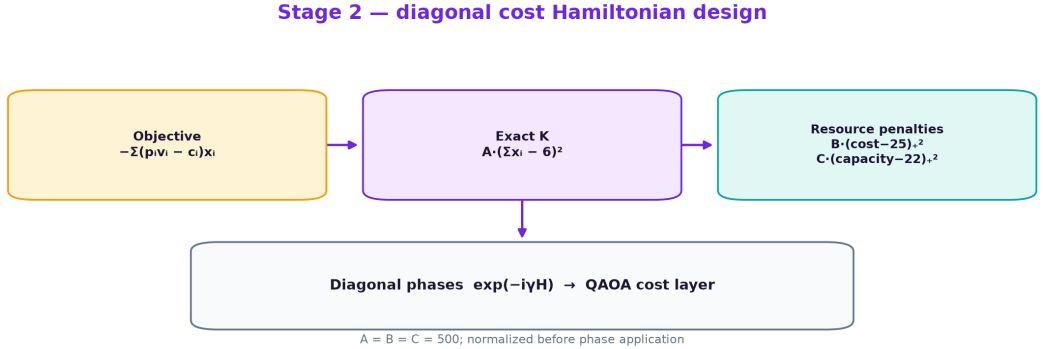
Continuous probabilities become candidates



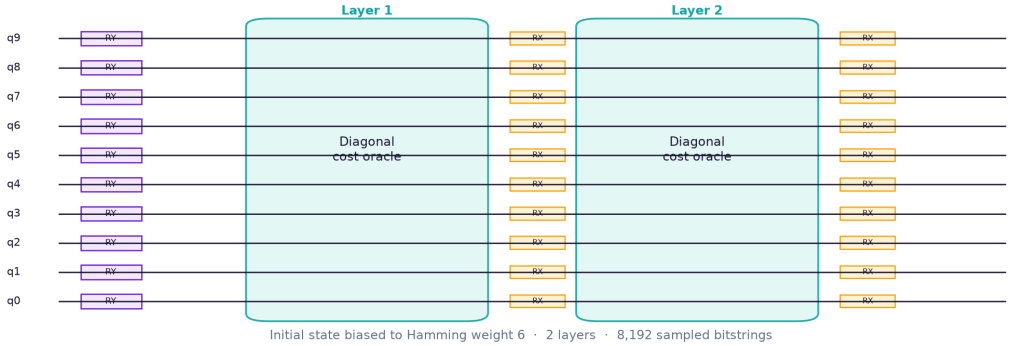
STAGE 2

Constraint-encoded quantum optimization

Penalty Hamiltonian

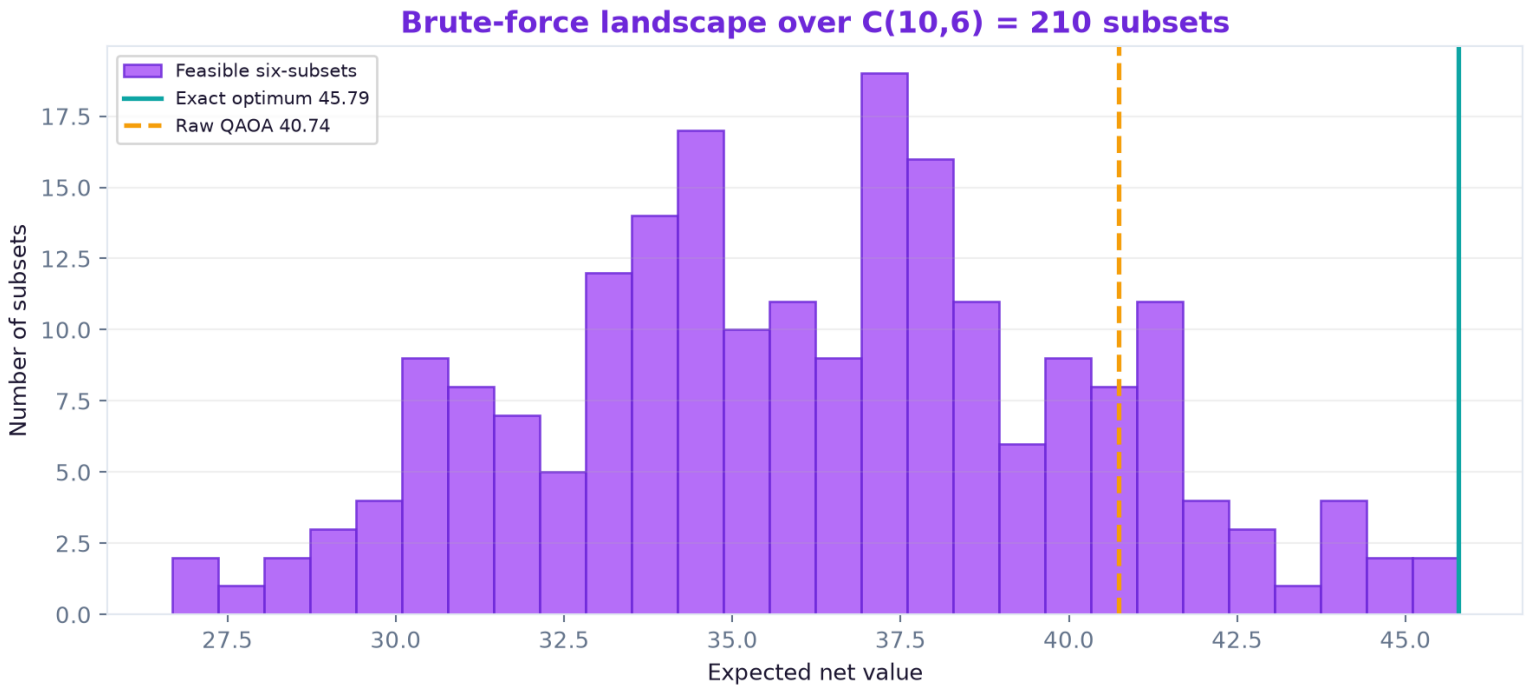


10-qubit QAOA



STAGE 2

Measured QAOA + zero-gap certificate



Certified decision

Exact objective 45.7874

Raw QAOA gap 5.0477

Final gap 0.0000

Budget 11 / 25

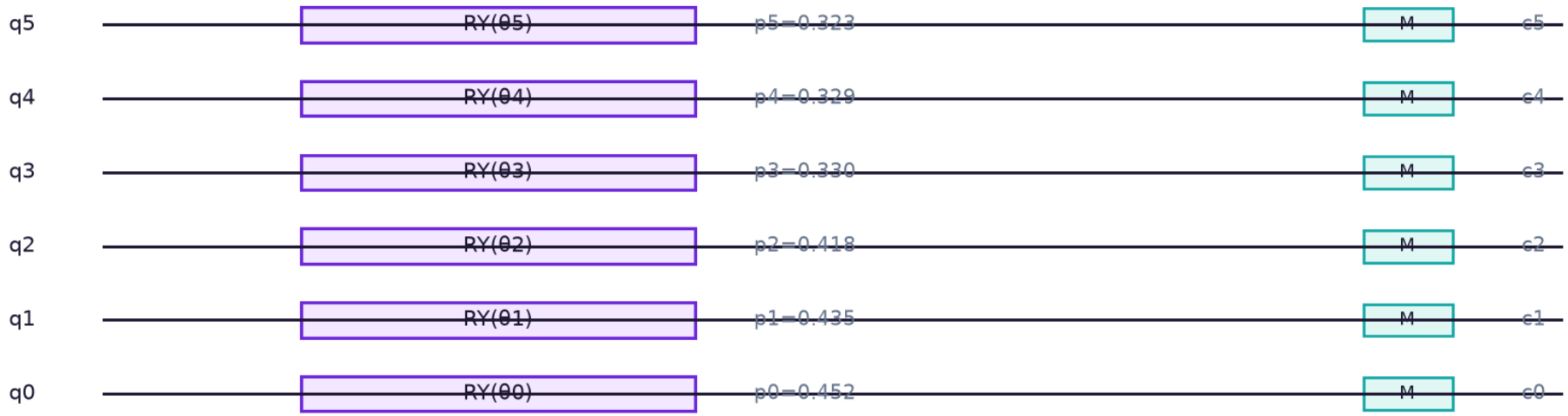
Capacity 16 / 22

210 subsets checked

STAGE 3

Six independent Bernoulli qubits

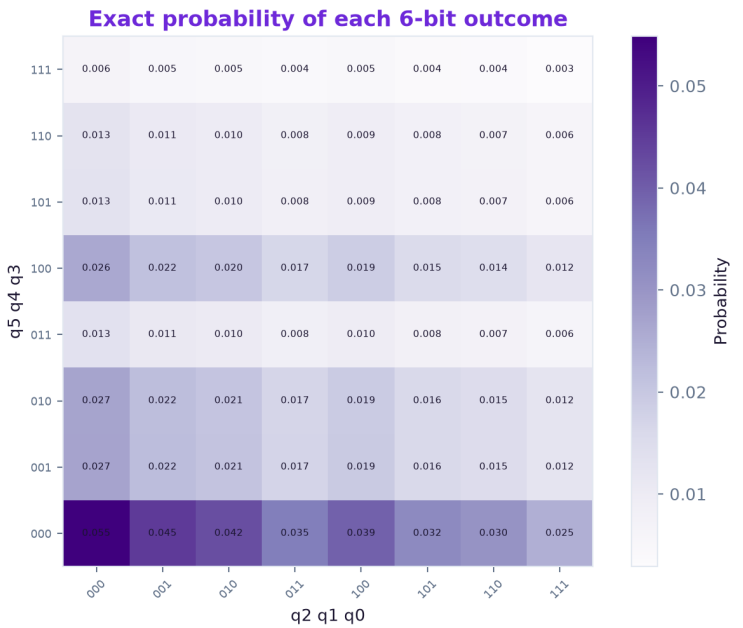
Six-qubit independent Bernoulli circuit

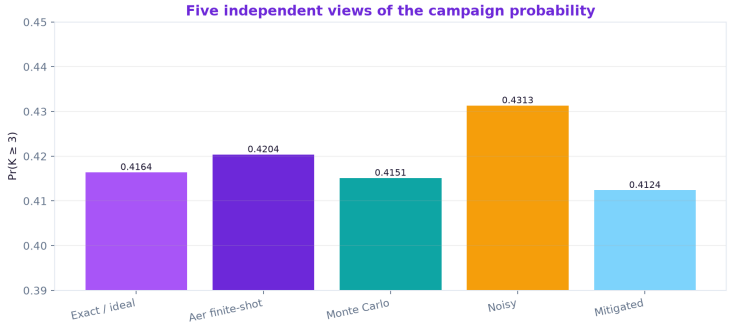


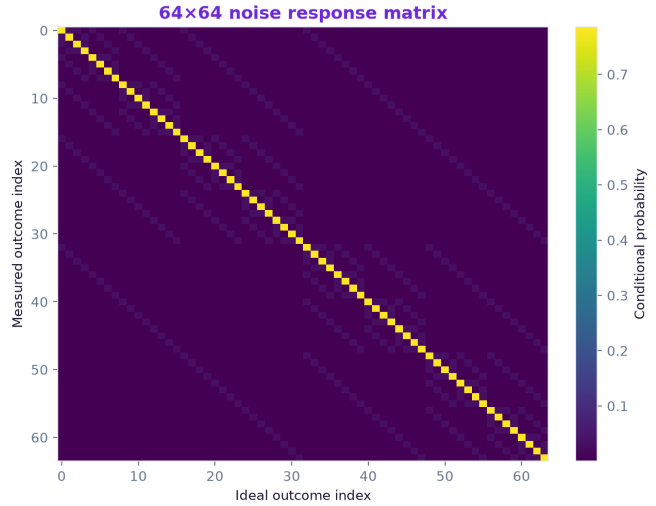
Success: Hamming weight of measured bitstring ≥ 3 · Qiskit prints c5…c0; q0 is rightmost

STAGE 3

Exact, finite-shot, noisy, and mitigated

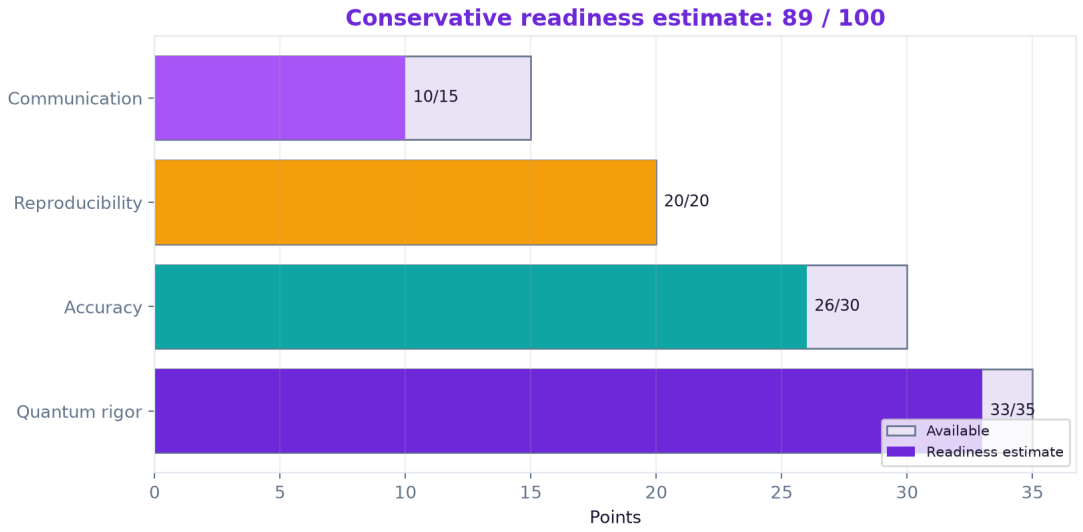






IMPACT

Auditable results and business value



Useful quantum computing is measurable and honest.

Campaign decision dashboard

Expected net value	Budget	Capacity	Pr(K ≥ 3)	Net / cost
45.7874	11 / 25	16 / 22	0.4164	4.1625×

Selected six: C0059 · C0170 · C0256 · C0327 · C0372 · C0374

Expected conversions 2.2870 · Expected deposit value 56.7874 · Total contact cost 11

Decision: contact only the certified six; retain the exact conversion probability as the campaign risk reference.